

# **TASK 3 – CODING PRACTICE (FULL DETAILS)**

## **Objective**

The objective of Task 3 is to improve problem-solving skills by practicing coding on an online platform like HackerRank and understanding logical thinking.

## **Platform Used**

HackerRank

Website: <https://www.hackerrank.com> 

## **Problem Selected**

Problem Name:  $3n + 1$  Problem (Collatz Sequence)

## **Problem Description**

The problem is based on a mathematical sequence:

Start with an integer  $n$

If  $n$  is even  $\rightarrow$  divide it by 2

If  $n$  is odd  $\rightarrow$  multiply by 3 and add 1

Repeat the process until  $n$  becomes 1

The total number of steps (including 1) is called the cycle length

# Logic Used

Take input value n

Initialize counter = 1

Repeat until n becomes 1:

If n is even  $\rightarrow n = n / 2$

Else  $\rightarrow n = 3 * n + 1$

Increase counter

Print total count

## Program Code (Python)

```
n = int(input())
```

```
count = 1
```

```
while n != 1:
```

```
    if n % 2 == 0:
```

```
        n = n // 2
```

```
    else:
```

```
        n = 3 * n + 1
```

```
    count += 1
```

```
print(count)
```

**Sample Input**

**22**

**Sample Output**

**16**

## **Output Explanation**

**For input 22, the sequence becomes:**

**22 → 11 → 34 → 17 → 52 → 26 → 13 → 40 → 20 → 10 → 5 → 16 → 8 → 4  
→ 2 → 1**

**Total steps = 16**

## **Proof of Work**

Python 3

```
1 n = int(input())
2 count =1
3 while n != 1:
4     if n% 2 ==0:
5         n=n//2
6     else:
7         n=3*n+1
8         count += 1
9 print(count)
```

Line: 7 Col: 15

Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Testcase 0

Congratulations, you passed the sample test case.

Click the **Submit Code** button to run your code against all the test cases.

Input (stdin)

22

# The 3n + 1 Problem

locked

Problem

Submissions

Submitted a few seconds ago • Score: 1.00

Status: Accepted

|  |              |  |              |  |              |
|--|--------------|--|--------------|--|--------------|
|  | Test Case #0 |  | Test Case #1 |  | Test Case #2 |
|  | Test Case #3 |  |              |  |              |

## Submitted Code

Language: Python 3

Open in editor

```
1 n = int(input())
2 count =1
3 while n != 1:
4     if n% 2 ==0:
5         n=n//2
6     else:
7         n=3*n+1
8         count += 1
9 print(count)
```

**Learned problem-solving techniques**

**Improved logical thinking**

**Understood loop and conditional statements**

**Gained hands-on experience with HackerRank platform**

## **Conclusion**

**Task 3 helped in developing coding skills and understanding how to solve real-world logical problems using programming. It improved confidence in writing and testing code.**

# **1. Title**

**Task 3: Digital Literacy Awareness Quiz using Google Forms**

# **2. Objective**

**The objective of this task is to create awareness about digital literacy and online safety using a Google Form quiz.**

# **3. Tools Used**

**Google Forms**

**Internet Browser**

**GitHub**

# **4. Description of Work**

**Example:**

**Created a Google Form**

**Added quiz questions**

**Enabled email collection**

**Shared form with users**

# **5. Topics Covered in Quiz**

**Digital Literacy**

Cyber Security

Strong Password

Phishing Awareness

Online Safety

## 6. Google Form Link

<https://docs.google.com/forms/d/e/1FAIpQLScp3GXatodoWnsxzl--h8-oqQ77PrxuJj-974ZiRq0Sul1aIA/viewform?usp=publish-editor>

## 7. Screenshots

← Preview mode ✓ Published [Copy responder link](#)

### Digital Literacy Awareness Quiz

\* Indicates required question

Email \*

☐ Record nirajyadav15867@gmail.com as the email to be included with my response

Q1. What is Digital Literacy?

☐ Ability to use digital devices

☐ Cooking skills

☐ Driving skills

☐ None

Q2. Which platform is used for coding practice?

☐ HackerRank

Q2. Which platform is used for coding practice?

- ☐ HackerRank
- ☐ Instagram
- ☐ WhatsApp
- ☐ YouTube

Q3. What is phishing?

Your answer

Q4. Which is a strong password?

- ☐ 123456
- ☐ password
- ☐ Abc@1234
- ☐ 111111



Q4. Which is a strong password?

- ☐ 123456
- ☐ password
- ☐ Abc@1234
- ☐ 111111

Q5. Why is online safety important?

Your answer

A copy of your responses will be emailed to nirajyadav15867@gmail.com.

Submit

[Clear form](#)

Never submit passwords through Google Forms.

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## **8. Conclusion**

**This task helped me understand how to create online forms and spread awareness about digital literacy.**